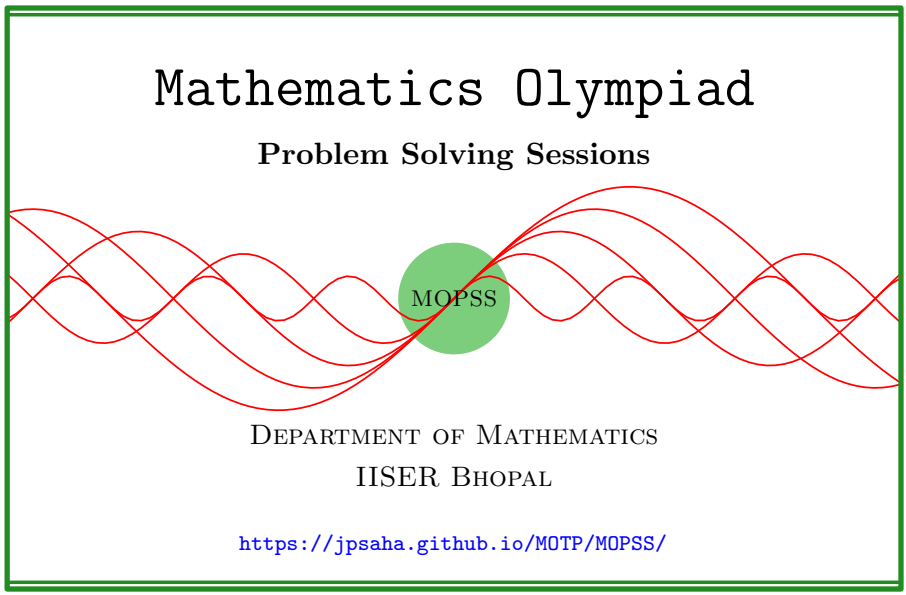
Integer divisibility

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Integer divisibility

Exercise 1. If $P(x)$ is a polynomial with integer coefficients and a, b are integers, then $P(a) - P(b)$ is a multiple of $a - b$.

> **Walkthrough** —
>
> **(a)** Show that it suffices to prove it for monomials.

Solution 1. ■

Exercise 2 (Flanders Mathematical Olympiad 2025 P3,). Consider the polynomial $P(x) = 1 + x + x^2 + ... + x^{2023} + x^{2024}$. What is the remainder polynomial when $P(x^{2025})$ is divided by $P(x)$?

> **Walkthrough** — Show that $P(x^{2025}) - 1$ is divisible by $P(x)$.

Solution 2. Note that $P(x^{2025}) - P(1)$ is divisible by $x^{2025} - 1$ and hence by $P(x)$. Therefore, the remainder polynomial when $P(x^{2025})$ is divided by $P(x)$ is equal to $P(1) = 2025$. ■

Example 3. Let $P(x)$ be a polynomial with integer coefficients such that $P(0), P(1)$ are odd. Show that $P(x)$ does not have any integer root.

Solution 3. If $P(x)$ admits an integer root α, then α divides $P(0)$ and $\alpha - 1$ divides $P(1)$, which shows that $\alpha, \alpha - 1$ are odd, which is impossible. ■

Exercise 4 (IOQM 2023 P12,). Let $P(x) = x^3 + ax^2 + bx + c$ be a polynomial where a, b, c are integers and c is odd. Let p_i be the value of $P(x)$ at $x = i$. Given that $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, find the value of $p_2 + 2p_1 - 3p_0$.

Summary — Using parity conditions, show that $p_1 + p_2 + p_3 \neq 0$. This gives $p_1 = p_2 = p_3$. This gives a system of equations in a and b, which can be solved to find a and b.

Walkthrough —

(a) Note that if $p_1 + p_2 + p_3 \neq 0$, then using the condition that p_1, p_2, p_3 are real, it follows that $p_1 = p_2 = p_3$.

(b) Note that

$$\begin{aligned} p_1 &= 1 + a + b + c, \\ p_2 &= 8 + 4a + 2b + c, \\ p_3 &= 27 + 9a + 3b + c, \end{aligned}$$

which gives that

$$p_1 + p_2 + p_3 = 36 + 14a + 6b + 3c.$$

Since c is an odd integer, it follows that[a] $p_1 + p_2 + p_3 \neq 0$. Consequently, $p_1 = p_2 = p_3$.

(c) This gives

$$3a + b = -7, 5a + b = -19,$$

and hence, $a = -6, b = 11$.

(d) This gives

$$\begin{aligned} p_2 + 2p_1 - 3p_0 &= 10 + 6a + 4b + 3c - 3c \\ &= 18. \end{aligned}$$

[a] Since c is odd, it follows that p_2 is odd. Note that $p_1 - p_3$ is even (Why? Does it follow that $f(1) - f(3)$ is divisible by 2 for any polynomial $f(x)$ with integer coefficients?), and hence, $p_1 + p_2 + p_3$ is odd.

Solution 4. Since $P(x)$ is a polynomial with integer coefficients, it follows that $p_1 - p_3$ is divisible by 2. Since c is odd, it follows that p_2 is odd. This shows that $p_1 + p_2 + p_3$ is odd, and hence, $p_1 + p_2 + p_3 \neq 0$. Using $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, we obtain

$$(p_1 + p_2 + p_3)(p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1) = 0.$$

This gives

$$p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1 = 0,$$

oe equivalently,

$$(p_1 - p_2)^2 + (p_2 - p_3)^2 + (p_3 - p_1)^2 = 0,$$

which implies that $p_1 = p_2 = p_3$. This shows that

$$1 + a + b + c = 8 + 4a + 2b + c = 27 + 9a + 3b + c,$$

which yields

$$3a+b=-7, 5a+b=-19.$$

This gives $a=-6, b=11$. It follows that

$$\begin{aligned} p_2+2p_1-3p_0 &= 10+6a+4b+3c-3c \\ &= 10-36+44 \\ &= 18. \end{aligned}$$

■

Exercise 5 (RMO 2016g P8,). **.** At some integer points a polynomial with integer coefficients take values $1, 2$ and 3. Prove that there exist not more than one integer at which the polynomial is equal to 5.

Solution 5. Denote the polynomial by $P(x)$. On the contrary, let us assume that there are at least two distinct integers where $P(x)$ takes the value 5.

Let a, b, c be integers such that

$$P(a)=1, P(b)=2, P(c)=3.$$

Note that $a-b$ divides $P(a)-P(b)$, $b-c$ divides $P(b)-P(c)$. It follows that $a-b=\pm 1, b-c=\pm 1$. Since a, b are of opposite parity, and so are the integers b, c, we obtain that a, c are of the same parity. Noting that $c-a$ divides $P(c)-P(a)=2$, it follows that $c-a=\pm 2$. Combining this with $a-b=\pm 1, b-c=\pm 1$, we get $a-b=b-c=1$ or $a-b=b-c=-1$.

This shows that $P(b-1)=1, P(b)=2, P(b+1)=3$ holds or $P(b+1)=1, P(b)=2, P(b-1)=3$ holds. Note that in the first case, the polynomial $R(x):=P(x-b)$ takes the values $1, 2, 3$ at the integers $-1, 0, 1$ respectively. In the second case, the polynomial $S(x)=P(-x+b)$ takes the values $1, 2, 3$ at the integers $-1, 0, 1$ respectively. This proves that there is a polynomial $Q(x)$ with integer coefficients which takes the values $1, 2, 3$ at $-1, 0, 1$ respectively.

From the hypothesis, it follows that there are distinct integers i, j such that $Q(i)=Q(j)=5$. Note that $i-1$ divides $Q(i)-Q(1)=2$, i divides $Q(i)-Q(0)=3$, $i+1$ divides $Q(i)-Q(-1)=4$. Since i divdes 3, we obtain $i=\pm 1, \pm 3$. Using $Q(-1)=1, Q(1)=3$, we get $i\neq -1, i\neq 1$. This gives $i=\pm 3$. Noting that $i-1$ divides 2, we obtain $i\neq -3$, and hence $i=3$. Similarly, it follows that $j=3$. ■

Example 6. Let $P(x)$ be a polynomial with integer coefficients such that $P(20), P(25)$ are of absolute value equal to 1. Show that $P(x)$ does not vanish at any integer.

Walkthrough —

(a) If $P(x)$ vanishes at an integer α, then $\alpha - 20$ divides $P(20)$ and $\alpha - 25$ divides $P(25)$.

Solution 6. On the contrary, let us assume that $P(x)$ vanishes at an integer α. Note that $\alpha - 20$ divides 1, and so does $\alpha - 25$. This shows that $\alpha - 20, \alpha - 25$ are of absolute value equal to 1. Applying triangle inequality, we obtain

$$5 \leq |\alpha - 20| + |\alpha - 25| \leq 2,$$

which is impossible. ■

Exercise 7 (USAMO 1974 P1,). Let a, b, and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

Walkthrough —

(a) Show that each of the integers $a - b, b - c, c - a$ is a multiple of the remaining two integers.

(b) Prove that this implies that a, b, c are equal.

Solution 7. Note that

$$a - b \mid P(a) - P(b) = b - c \mid P(b) - P(c) \mid c - a \mid P(c) - P(a) = a - b.$$

Consequently, the integers $a - b, b - c, c - a$ are of the same absolute value. Denote their absolute value by k. Note that their sum is zero. However, the sum is equal to mk, for some $m \in \{\pm 1, \pm 3\}$. Hence, k is equal to zero. This yields that $a = b = c$. ■

Here is a more general result.

Example 8. Let $P(x)$ be a polynomial with integer coefficients, and let n be an odd positive integer. Suppose that $x_1, x_2, \ldots, x_n$ is a sequence of integers such that $x_2 = P(x_1), x_3 = P(x_2), \ldots, x_n = P(x_{n-1})$, and $x_1 = P(x_n)$. Prove that all the x_i's are equal.

Walkthrough — Show that

$$a_1 - a_2 \mid a_2 - a_3 \mid a_3 - a_4 \mid \cdots \mid a_n - a_1 \mid a_1 - a_2.$$

Note that sum of these differences is an odd multiple of their absolute value.

Exercise 9 (Tournament of Towns Spring 2014, Senior A Level P4, by G. K. Zhukov)**.** In the plane, the points with integer coordinates (x, y) satisfying $0 \leq y \leq 10$ are marked. Consider a polynomial of degree 20 with integer coefficients. Determine the maximum possible number of marked points which can lie on its graph.

Walkthrough —

(a)

Solution 8. Note that the polynomial

$$(x-1)(x-2)(x-3)\dots(x-20)$$

of degree 20 has integer roots. Let us prove that the graph of no polynomial of degree 20 with integer coefficients passes through more than 20 marked points.

Claim — Let $P(x)$ be a polynomial of degree 20 with integer coefficients. No more than 20 marked points lie on the graph of $P(x)$.

Proof of the Claim. On the contrary, let us assume that there are integers $x_1 < x_2 < \cdots < x_{21}$ such that

$$0 \leq P(x_i) \leq 10$$

holds for all $1 \leq i \leq 21$. For any integer $1 \leq i \leq 10$, the inequality $x_{21} - x_i \geq 11$ holds, and using that $x_{21} - x_i$ divides the integer $P(x_{21}) - P(x_i)$, which lies in $[-10, 10]$, it follows that $P(x_{21}) = P(x_i)$. Similarly, for any integer $12 \leq i \leq 21$, it follows that $P(x_1) = P(x_i)$. This shows that $P(x_i) = P(x_1)$ for any integer $i \in \{1, 2, \dots, 10\} \cup \{12, 13, \dots, 21\}$. Since $P(x)$ is a polynomial of degree 20, it follows that

$$P(x) - P(x_1) = c(x-x_1)(x-x_2)\dots(x-x_{10})(x-x_{12})(x-x_{13})\dots(x-x_{21})$$

holds for some nonzero integer a. This yields that

$$|P(x_{11}) - P(x_1)| \geq (10!)^2,$$

which is impossible. □

This proves that the maximum possible number of marked points which can lie on the graph of a polynomial of degree 20 with integer coefficients is equal to 20. ■

Exercise 10 (RMO 2025a P6,). Let $p(x)$ be a nonconstant polynomial with integer coefficients, and let $n \geq 2$ be an integer such that no term of the sequence

$$p(0), p(p(0)), p(p(p(0))), \ldots$$

is divisible by n. Show that there exist integers a, b such that $0 \leq a < b \leq n-1$ and n divides $p(b) - p(a)$.

Walkthrough —

(a) Apply the pigeonhole principle to show that some two terms of the given sequence are congruent modulo n.

(b) Let i and j be positive integers such that $i < j$ and the i-th and j-th terms of the sequence are congruent modulo n.

(c) Reduce the arguments of these two terms modulo n to obtain integers a and b such that $0 \leq a, b \leq n-1$ and n divides $p(b) - p(a)$.

(d) Show that $a \neq b$. It could be useful to assume that the integer i as above is the smallest with this property.

Solution 9. Define the sequence

$$p^{(0)}(0), p^{(1)}(0), p^{(2)}(0), p^{(3)}(0), \ldots$$

by

$$p^{(n)}(0) \coloneqq \begin{cases} 0 & \text{if } n = 0, \\ p(p^{(n-1)}(0)) & \text{if } n \geq 1. \end{cases}$$

By the pigeonhole principle, some two among the following $n+1$ integers

$$p^{(1)}(0), p^{(2)}(0), \ldots, p^{(n+1)}(0)$$

are congruent modulo n. Let i denote the smallest positive integer such that there exists a positive integer $j > i$ with

$$p^{(i)}(0) \equiv p^{(j)}(0) \pmod{n}.$$

Let $0 \leq a \leq n-1$ and $0 \leq b \leq n-1$ be integers such that

$$a \equiv p^{(i-1)}(0) \pmod{n}, \quad b \equiv p^{(j-1)}(0) \pmod{n}.$$

Note that n divides $p(b) - p(a)$. We claim that $a \neq b$. Suppose, for the sake of contradiction, that $a = b$. It follows that

$$p^{(i-1)}(0) \equiv p^{(j-1)}(0) \pmod{n}.$$

Using $j - 1 > i - 1$ and the minimality of i, we obtain $i = 1$. This implies that

$$p^{(0)}(0) \equiv p^{(j-1)}(0) \pmod{n}.$$

Since $p^{(0)}(0) = 0$ and $j - 1 > 0$, we have that n divides $p^{(j-1)}(0)$, contradicting the given condition. This implies that $a \neq b$.

Reordering if necessary, we may assume that $a < b$. Consequently, there exist integers $0 \leq a < b \leq n - 1$ such that n divides $p(b) - p(a)$. ■

Lemma 1

Let P be a polynomial with integer coefficients. Suppose a is an integer and k is a positive integer such that $P^k(a) = a$, where P^k denotes the k-fold composite map from $\mathbb{Z} \to \mathbb{Z}$. Show that $P^2(a) = a$.

Proof. Let ℓ denote the smallest positive integer such that $P^\ell(a) = a$. If $\ell = 1$ or $\ell = 2$, then we are done. Henceforth, we assume that $\ell \geq 3$.

Note that

$$P(a) - a \mid P^2(a) - P(a) \mid \cdots \mid P^\ell(a) - P^{\ell-1}(a) = a - P^{\ell-1}(a) \mid P(a) - a.$$

Since $a - P^{\ell-1}(a)$ is nonzero, it follows that the above differences are nonzero. Consequently, for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = \pm(P^i(a) - P^{i-1}(a)).$$

If $P^{i+1}(a) = P^{i-1}(a)$ holds for some $1 \leq i \leq \ell$, then applying $P^{\ell-i+1}$ to both sides, we obtain $P^2(a) = a$, which contradicts the assumption that $\ell \geq 3$. It follows that for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = P(a) - a$$

holds, which implies that

$$\sum_{i=0}^{\ell-1} (P^{i+1}(a) - P^i(a)) = \ell(P(a) - a).$$

This gives $P(a) = a$, which contradicts the assumption that $\ell \geq 3$. This completes the proof. □

Exercise 11 (IMO 2006 P5, by Dan Schwarz, Romania)**.** Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients, and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\ldots P(P(x))\ldots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

Walkthrough —

(a) Does the above lemma help?

The content posted here and at this blog by Evan Chen are quite useful.

Solution 10. By the above lemma, it reduces to considering the case $Q(x) = P^2(x)$.

Suppose Q has more than n fixed points. Since P is not linear, it follows that P cannot have n fixed points, and hence not all the fixed points of Q are fixed points of P. Let b be a non-fixed point of P, and $Q(b) = b$. Suppose a be a fixed point of Q, other than b.

Let us first consider the case that $P(a) \neq a$. Note that

$$P(b) - a \mid P(a) - b \mid P(b) - a$$

holds, and

$$a - b \mid P(a) - P(b) \mid a - b$$

holds too. This yields that

$$|P(a) - b| = |P(b) - a|, \quad |P(a) - P(b)| = |a - b|.$$

If

$$P(a) - b = a - P(b), \text{ and } P(a) - P(b) = a - b$$

hold, then b would be a fixed point of P. It follows that at least one of

$$P(a) - b = -(a - P(b)), P(a) - P(b) = -(a - b)$$

holds. Consequently, we obtain

$$P(a) + a = P(b) + b.$$

Next, let us consider the case that $P(a) = a$. Note that

$$P(b) - a \mid b - a \mid P(b) - a.$$

Since b is not a fixed point for P, it follows that

$$P(b) - a = a - b,$$

which yields

$$P(a) + a = P(b) + b.$$

This proves that all the roots of $Q(x) = x$ are the roots of $P(x) + x = P(b) + b$. Since $P(x)$ has degree $n > 1$, it follows that the polynomial $P(x) + x - P(b) - b$ is of degree n, and it has more than n roots, which is impossible.

Hence, there are at most n integers t such that $Q(t) = t$ holds. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)